International Journal of Human–Computer Interaction

ISSN: 1044-7318 (Print) 1532-7590 (Online) Journal homepage: www.tandfonline.com/journals/hihc20

Asking AI to Eat Its Own Cooking: Artificial Experience vs. Hypothetical Experience in Playtesting

Yoones A. Sekhavat

INTERNATIONAL JOURNAL OF HUMAN–COMPUTER INTERACTION
https://doi.org/10.1080/10447318.2025.2587239

Taylor & Francis
Taylor & Francis Group

Asking AI to Eat Its Own Cooking: Artificial Experience vs. Hypothetical Experience in Playtesting

Yoones A. Sekhavat

Department of Mathematics & Computer Science, Modern College of Business and Science, Muscat, Oman

ABSTRACT
Game design is an iterative process that relies on prototyping and playtesting to refine mechanics and enhance player experience. Although Artificial Intelligence (AI) tools increasingly assist in content creation, early gameplay evaluation still depends on time-intensive human testing. This study investigates whether AI can act as a proxy for player feedback through two approaches: Artificial Intelligence Experience (AX) and Hypothetical Experience (HX). AX uses AI agents to play the game and generate feedback based on player-centric metrics, while HX analyzes the Game Design Document (GDD) to simulate hypothetical player responses. Both methods were applied to a casual game prototype to evaluate their usefulness in identifying design issues and suggesting improvements. Results show that AX provides actionable insights from gameplay, and HX supports conceptual coherence analysis, offering low-cost, rapid feedback to accelerate early-stage iteration without replacing traditional UX testing.

KEYWRODS
Player experience; AI-based playtesting; user experience; video game design; game prototyping

1. Introduction

Game development is fundamentally driven by game design, which is an inherently complex and iterative process aimed at enhancing the overall quality of gameplay (Meşe et al., 2025; Schell, 2008). To refine their ideas and mechanics, designers rely heavily on rapid prototyping and playtesting, allowing them to experiment quickly and gather feedback during early development stages (Fullerton, 2024; Jiang et al., 2022). However, delivering high-quality results within tight development timelines and limited budgets remains a significant challenge.

Artificial Intelligence (AI) has become increasingly prominent in supporting game designers. Through domain-specific languages, procedural content generation, and intelligent tools, AI helps automate various design processes (Cook, 2020). These tools have been applied to different aspects of games such as mechanics (Kreminski et al., 2020) levels (Liapis et al., 2013), missions, narratives, and virtual environments (Sekhavat et al., 2025; Sekhavat & Tabatabaei, 2025). Using mixed-initiative approaches assists designers in navigating complex design spaces, enabling a collaborative workflow where both the AI and the designer contribute to content creation and decision-making (van Rozen, 2025). By leveraging AI, designers can define rules, balance gameplay, and model player behavior with a high degree of precision.

Despite these advancements, transforming complex designs into playable systems remains time-consuming, often resulting in the loss of design intent. As prototypes evolve, they can become too complex to manage effectively. As Raph Koster once remarked, “a game design should fit on the back of a napkin” (Koster, 2013), but in reality, making prototypes playable to collect meaningful feedback is still a major hurdle. While recent developments in Large Language Models (LLMs), such as GPT-3, have lowered the barrier for quickly generating playable prototypes (Jiang et al., 2022), these systems still require player interaction and sufficient playtime to produce valuable gameplay data (Colado et al., 2023). Moreover, current prototyping tools often lack integrated support for continuous feedback, real-

CONTACT Yoones A. Sekhavat yoones.sekhavat@mcbs.edu.om Department of Mathematics & Computer Science, Modern College of Business and Science, Bawshar Street, Way No. 273, Building No. 259, Muscat, Oman

time behavior tracking, and evaluation mechanisms (Cook, 2020). As a result, iterating on game ideas is slow, forming mental models about cause-effect relationships is difficult, and the learning curve for programming game behaviors can be steep and frustrating.

AI technologies are rapidly entering various domains of game production, particularly with the widespread use of LLMs. These tools are increasingly used to speed up content generation and streamline early-stage prototyping. However, the ultimate goal of game design remains the same: to deliver engaging experiences for players. Traditionally, evaluating the player experience involves conducting user studies with human participants, which is an essential but resource-intensive step, especially in early development.

This article investigates whether AI can serve as a proxy for human players in evaluating gameplay during the early stages of design. Specifically, we introduce the concept of Artificial Intelligence Experience (AX) and Hypothetical Experience (HX), which refers to using AI agents to evaluate the playability and quality of casual games:

1. Artificial Intelligence Experience (AX): Having an AI agent play the game and report its experience based on predefined metrics of player experience.
2. Hypothetical Experience (HX): Asking an AI to analyze the Game Design Document (GDD) and simulate feedback as if it were a hypothetical player.

We apply both techniques to a case study involving a simple casual game prototype. Our findings suggest that evaluating the game using AX and HX approaches can lead to meaningful design improvements, helping designers iterate more efficiently in early phases when human playtesting may be costly or impractical. We also show that AX can suggest more practical improvement based on real game play data. Importantly, we do not position AX as a replacement for traditional user experience (UX) evaluations. Rather, it is an early-stage augmentation tool that provides fast, low-cost insights into gameplay quality to guide iterative development.

2. Background and related work

Innovations in AI are dramatically altering approaches to designing games. AI is no longer limited to content generation; it now plays a central role in improving design efficiency and streamlining development workflows (Moon et al., 2024). By enabling the creation of adaptive virtual agents and dynamic game environments, AI allows games to adjust in real time to players' skill levels and behaviors (Yang et al., 2025). For example, generative AI can produce quests, characters, or levels tailored to individual preferences, resulting in more personalized, replayable experiences. These systems analyze player input and performance to evolve game content dynamically, enhancing engagement and responsiveness (Barthet et al., 2022).

Beyond content adaptation, AI supports the design of open-ended game environments that present players with diverse challenges, spatial layouts, and interaction possibilities (Lv, 2023). Advances in character and agent design have led to more complex and reactive personalities, enriching social interactions and narrative depth within games (Barthet et al., 2022; Kumaran et al., 2023). Interactive artworks can engage audiences by subtly responding to personal attributes and movements, encouraging reflection on social interactions and human behavior (Sekhavat & Tabatabaei, 2025). This evolution moves games away from linear, fixed experiences toward fluid systems that respond intelligently to player actions (Anjum et al., 2024; Fui-Hoon Nah et al., 2023). Such adaptive systems rely on probabilistic and statistical techniques to analyze large volumes of unstructured data, generating outputs that emulate human decision-making and creativity (Baidoo-Anu & Ansah, 2023).

AI also accelerates the practical aspects of game development by automating tasks from early prototyping to user testing. Tools like Ludo and Meshy AI can auto-generate game assets and procedural elements, allowing designers to concentrate on high-level creativity while offloading repetitive tasks to machines (Ratican & Hutson, 2024). Large Language Models (LLMs), such as OpenAI's Codex, further enhance development by translating natural language into executable code, enabling the rapid implementation of game mechanics, environments, and character interactions with minimal manual coding

(Weber, 2025). Emerging multimodal models, including Google's Gemini, extend these capabilities by integrating textual, auditory, and visual inputs, supporting the creation of rich, immersive experiences (Sekhavat & Zarei, 2018) that respond to multiple data streams.

Collectively, these advances position AI as a central enabler of procedural generation, adaptive gameplay, and enhanced player interactivity. By combining real-time responsiveness with automated content creation, AI is reshaping not only how games are developed but also how players experience and engage with them, setting the stage for increasingly personalized and dynamic game worlds.

2.1. The role of playtesting in game development

Playtesting represents a structured and essential phase in game development where player feedback is systematically collected and analyzed. Its goal is to assess how closely the game aligns with the designer's intent and the target audience's expectations (Fullerton, 2024). Rather than being a single method, playtesting often involves a blend of user research techniques, such as observation, interviews, surveys, and even physiological measurements. Each approach offers different insights, and selecting the right combination is a critical task for user researchers when planning effective studies.

Because playtesting varies widely in scope and implementation, there is no universally accepted definition, and its interpretation often depends on the context of a game's development cycle (Mirza-Babaei et al., 2020). Most professional games undergo some level of testing either in-house or through external partners; for example, Halo 3 involved over 3,000 hr of gameplay from more than 600 participants at one of the world's most advanced playtesting labs, illustrating the scale often required to capture diverse player behaviors and ensure robust design feedback. Conducting such large-scale tests can be resource-intensive and time-consuming, which our work addresses by introducing Artificial Intelligence Experience (AX) and Hypothetical Experience (HX). These AI-based approaches simulate player behavior and provide experiential feedback, offering a scalable alternative for early-stage evaluation of game design decisions.

Despite differing methodologies, the underlying objective of all playtesting activities is the same: to gather actionable insights from players that help enhance the game experience. The process usually begins with the formulation of clear research questions and goals. Researchers then gather behavioral data (e.g., "can players achieve this objective?") and attitudinal data (e.g., "do players enjoy this aspect?") through observational techniques, note-taking, and follow-up methods like interviews or questionnaires (Mirza-Babaei et al., 2020). Studies indicate that enjoyment in games depends on complex interactions between design elements rather than any single factor (Caroux & Pujol, 2024). The resulting findings are analyzed and presented in a report that guides design revisions and decision-making (Mirza-Babaei et al., 2016).

The scale and quality of playtesting often depend on a studio's resources. Larger companies may have internal user research teams and dedicated facilities, ensuring a consistent feedback loop throughout development, while smaller studios may rely on outsourced testing or skip formal playtesting due to financial constraints. These limitations are compounded by the scarce availability of public data on the return on investment (ROI) of playtesting and its correlation with game quality, and by the fact that non-research roles in game development, such as programmers or artists, may not be well-versed in testing methodologies (Mirza-Babaei et al., 2016b). Effective playtesting requires informed decisions, methodological rigor, and clear communication among all stakeholders, including design, development, production, and marketing teams (Choi et al., 2016). Failure to apply proper techniques can lead to poor feedback quality, inefficient use of resources, and ultimately lower-quality games, increasing market failure rates and challenging the justification of structured testing practices (Isbister & Schaffer, 2008). In this context, AI can play an important role in simulating player behavior, providing scalable early-stage feedback to inform design decisions even when traditional playtesting is limited.

2.2. AI in playtesting

Artificial intelligence is increasingly integral to all stages of user experience (UX) design in games, from identifying user needs to evaluating in-game experiences. During the evaluation phase, AI techniques are commonly applied to assess UX through a range of cognitive, emotional, and behavioral indicators, which can be extracted from multiple data sources, including facial expressions, physiological measurements, and gameplay activity logs (Luo, 2025; Schoop et al., 2022). AI-driven systems can automatically detect usability issues by analyzing user testing recordings, reviews, and behavioral patterns. Additional methods, such as sentiment analysis, speech recognition, and the creation of simulated user personas, further enhance this evaluation process (Fan et al., 2022).

Generative AI tools are also increasingly used in qualitative UX research. For example, they can assist in designing interview questions, simulating user-researcher interactions, or drafting survey prompts, thereby improving the efficiency and consistency of study planning and execution (Takaffoli et al., 2024). Earlier in the UX workflow, AI can capture and interpret user behavior in naturalistic settings by analyzing sensor data, interaction logs, and environmental cues, helping identify patterns of engagement, potential frustration points, or triggers for disengagement (Luo et al., 2024). One particularly impactful application of AI is persona generation. Predictive models leverage quantitative behavioral data to construct detailed user personas (Hu et al., 2023), while generative AI tools, such as ChatGPT, can create rich textual and visual narratives to represent these personas more vividly (Huang et al., 2024). By providing game designers with a comprehensive understanding of their audience, these AI-generated personas enable more informed and targeted design decisions.

Findings indicate that competing with AI in prediction tasks enhances fans' curiosity and engagement during game experiences (Jang et al., 2025). However, the effectiveness of AI in UX evaluation and playtesting depends heavily on the availability of gameplay data. Without sufficient user interaction data, AI systems have limited ability to provide meaningful feedback or guide design improvements. To overcome this limitation, we propose methods that allow AI to generate simulated playtest data, effectively acting as artificial and hypothetical players. This approach enables early feedback and iterative design improvements before involving human participants, supporting a more efficient and scalable design process during the initial development stages.

3. Materials and methods

We propose and compare two techniques for evaluating player experience during the game design process. The first technique, which we refer to as Artificial Experience (AX), involves asking an AI agent to play the game and report its experience using a predefined set of player experience metrics. These metrics comprise 16 measures grouped into four categories: Gameplay Experience, Challenge & Mastery, Usability & Learning, and Esthetics & Feedback, as outlined in Table 1. After completing its gameplay session, the AI generates evaluative feedback based on these categories, simulating how a human player might reflect on the game. This approach allows for a rapid and scalable evaluation of the player experience during early prototyping.

The second technique, which we call Hypothetical Experience (HX), involves prompting the AI to analyze the game design document and provide feedback as if it were a hypothetical player experiencing the game. Rather than interacting directly with the game, the AI simulates how a player might perceive and interpret the game based solely on its description, mechanics, goals, and structure. This method is particularly useful during the early stages of development, when a playable version of the game may not yet be available. HX offers designers a way to anticipate user reactions and identify potential issues before implementation.

We apply and evaluate both techniques in the context of a casual game case study. Our findings demonstrate how using AX and HX to assess and refine the game can enhance design quality.

Table 1. Player experience metrics used by AI to evaluate the prototypes.

Gameplay experience	Challenge & mastery	Usability & learning	Esthetics & feedback
Fun/Enjoyment	Challenge level	Clarity of goals and rules	Visual appeal
Engagement/Involvement	Competence/Mastery	Ease of learning	Audio appeal
Flow/Immersion	Sense of progress	User interface clarity	Reward satisfaction
Replayability	Stress/Frustration level	Responsiveness of controls	Pacing

Furthermore, we highlight the differences between the two approaches in the type and depth of feedback they provide, and how each contributes to improving the overall game design process.

3.1. Prototyping, playtesting and feedback loop

The overall procedure of prototyping, playtesting, feedback and revising the game is illustrated in Figure 1. As illustrated, the process begins with the ideation of a simple casual game. To implement a fully AI-driven pipeline for both generating and evaluating prototypes, the initial game design document was created using GPT − 4 turbo. To maintain full control over game variables and mechanics, a casual game was chosen as the case study. This decision allowed for a more focused experimentation by narrowing the scope of the game environment and mechanics. Specifically, the AI was prompted to generate a basic concept for a casual game in which both players and objects are represented as circles. No further details were provided intentionally, allowing the AI to rely solely on its own creativity without human input for the initial version.

The prompt requested a comprehensive outline covering game mechanics, the gameplay loop, game objects, rules, player-object interactions, environment, player dynamics, reward system, feedback mechanisms, and the scoring system. The game prototype was then developed using Cursor (version 1.0), an AI-powered code editor designed to enhance developer productivity through integrated AI features. The AI-generated design document was fed into Cursor, which was tasked with producing a playable HTML5 game based on the provided specifications.

3.2. Feedback loop using artificial experience (AX)

To implement and evaluate the Artificial Experience (AX) loop, we employed AI agents as simulated participants in the study. Each AI agent was programmed to autonomously play the game for a 30-second session immediately after launch, during which detailed gameplay data was recorded for analysis. Upon completing a playthrough, the AI agent generated a self-assessment of its gameplay experience based on 16 player experience measures, organized into four categories: Gameplay Experience, Challenge & Mastery, Usability & Learning, and Esthetics & Feedback, as outlined in Table 1. Simulated participants effectively act as stand-ins for human players, providing structured feedback that supports early-stage evaluation and informs iterative design decisions without relying on large-scale human testing. For each item in the evaluation table, the AI was asked to assign a score on

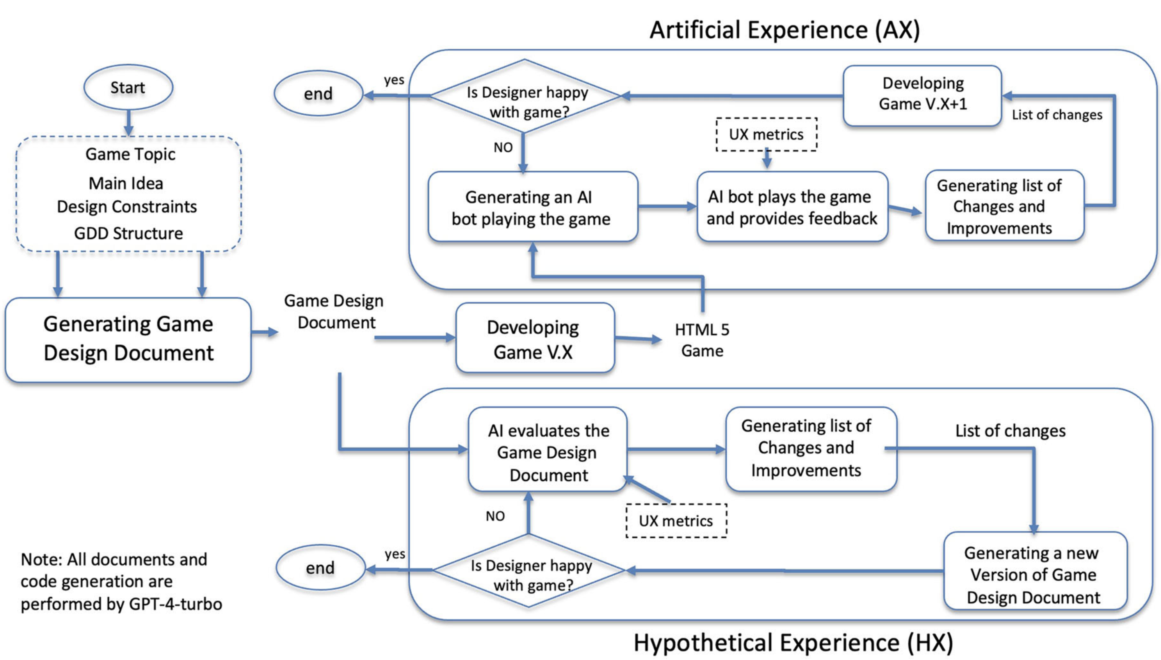

Figure 1. Research methodology and procedural framework.

a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), accompanied by a brief explanation justifying the given score. This approach enables both quantitative and qualitative insights into the AI's perceived experience.

Following the evaluation, Cursor was further prompted to analyze the AX results and propose potential improvements to enhance the overall artificial experience in future iterations of the game design. This closed-loop process simulates reflective assessment and iterative refinement typical in human-centered game development, but within a fully AI-driven framework.

3.3. Feedback loop using hypothetical experience (HX)

In the second approach called Hypothetical Experience (HX), AI agents acted as simulated participants to assess the game without interacting with a working prototype. Unlike the Artificial Experience (AX) method, where the AI plays the implemented game directly, this approach relies solely on the game design document (GDD) to generate feedback.

Using the original GDD introduced earlier, we instructed GPT – 4 Turbo to imagine playing the game as a hypothetical player. After this simulated gameplay, the AI-generated hypothetical player evaluated its experience using the same 16 player experience measures described previously. This approach allows early-stage assessment of design concepts and potential gameplay issues, providing structured feedback without requiring an operational game prototype. As in the AX approach, the AI was asked to assign a rating for each measure using a 7-point Likert scale, along with a concise explanation justifying each score. This approach enabled a structured, text-based evaluation of gameplay experience, allowing us to compare the reliability and richness of feedback derived from imagined play versus real-time interaction.

4. Results: The case study of CircleDash game

The first version of the Game Design Document for *CircleDash* game generated by AI is shown in Table 2. As evident from the design document, the game is a minimalist concept centered around collecting circular objects on the screen to earn points. It features simple yet engaging mechanics based on one-touch input. Players tap or click to make a controlled circle dash forward within a minimalist, single-screen arena. The core loop involves dashing to collect point-giving circles while avoiding red enemy circles that move along set paths. Players bounce off walls and obstacles, using timing and reflexes to navigate the space strategically. Collisions with enemies end the round, while collectibles increase the score and unlock cosmetic rewards. A snapshot of the game prototype created using

Table 2. The game design Document (GDD 1.0) generated for CircleDash game at the beginning by GPT-4.

Category	Details	Category	Details
Game mechanics	• Tap/click to dash forward • Bounce off walls/obstacles • Collect circles for points • Avoid red enemies	Environment	• Single-screen minimalist arena • Light background with solid-line boundaries • Randomized enemies and collectibles
Gameplay loop	• Dash • Collect • Avoid • Collision ends round • Earn rewards, • restart	Player Dynamics	• One-touch dash input • Timing and reflexes essential • Anticipate enemy paths • Plan movement strategically
Game objects	• Player Circle (controlled) • Enemy Circles (set paths) • Collectibles (points) • Walls (boundaries)	Reward System	• +1 point per collectible • Unlock cosmetics at 10/20/50 points • Daily bonus collectible
Rules	• Touching collectibles to get points • Touching enemies is game over • Bounce off walls • Movement only on tap	Feedback	• Visual: particles, screen shake • Audio: "Ping," "Thud," "Crash" • Haptics: vibration on hit/death (mobile)
Interactions	• Player–Collectible: +1 point • Player–Enemy: Game over • Player–Wall: Bounce • Enemy–Wall: Bounce	Scoring System	• +1 Point per collectible • Show high/last score after round • Bonus +1 for every 10s survived

Cursor 1.0 based on this design is shown in Figure 2. We refer to this version as *CircleDash 1.0* for reference.

4.1. Artificial experience (AX): Round 1

In the first round, we instructed Cursor to generate an AI player that simulates playing the CircleDash 1.0 game for 30 s. At the end of the session, the AI player was prompted to evaluate its gameplay experience using the 16 predefined player experience measures. The results of this evaluation, along with the AI's rationale for each score, are presented in Table 3, which hare notable. The AI bot identified several key issues that negatively affect the player experience. Cursor was also prompted to provide brief justifications for each score, which are presented in the explanation column for additional context. The evaluation indicates that the game offers a highly enjoyable and immersive experience, with top scores in fun, engagement, flow, and clarity of goals, suggesting that the gameplay is intuitive, consistent, and rewarding. Players are able to quickly understand the mechanics and experience a strong sense of progress and mastery.

However, the game received a score of 0 in both challenge level and stress/frustration, indicating a lack of difficulty and emotional tension. This may diminish long-term engagement and reduce the depth of the experience. Other aspects such as replayability, control responsiveness, and UI clarity were rated moderately high, reflecting solid design but leaving room for improvement. Esthetic elements like visual and audio appeal were rated as adequate, though they could benefit from more dynamic feedback and variety.

We asked Cursor to analyze the AX results and suggest improvements to enhance the overall artificial experience (AX) scores. The results of this analysis, along with the corresponding game design

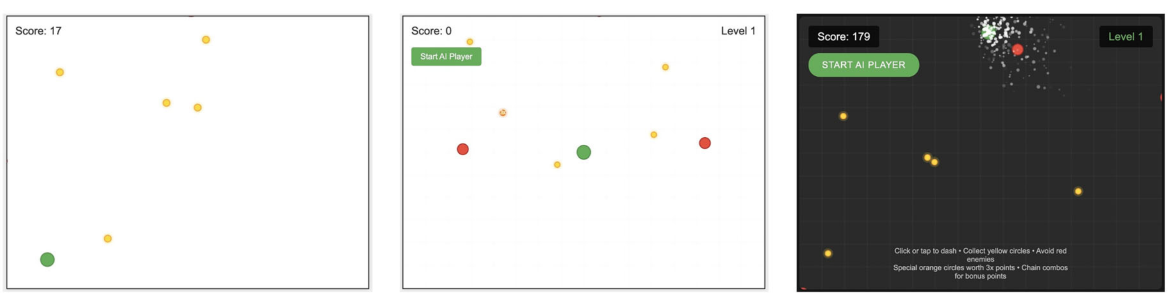

Figure 2. Screenshots of CircleDash 1.0 (left, https://sekhavat.itch.io/circledash-v1), CircleDash 2.0 (center, https://sekhavat.itch.io/circledash-v2), and CircleDash 3.0 (right, https://sekhavat.itch.io/circledash-v3), along with links to play and test the games online.

Table 3. Evaluation outcomes of player experience in CircleDash 1.0 and CircleDash 2.0, including AI-provided reasons for each score across all experience measures.

Playing experience aspect	Score version 1.0	Reasons behind the score by AI for CircleDash 1.0	Score version 2.0	Reasons behind the score by AI for CircleDash 2.0
Fun/Enjoyment	7	Points gathered and survival time	7	63 Items collected, 267 dodges
Engagement/Involvement	7	Continuous action and decisions	7	60.1 Decisions per second
Flow/Immersion	7	Smooth movement and collection	7	100% Time in motion
Challenge level	0	Few near-misses, low threat	4	6 Enemies, 9 close calls
Competence/Mastery	7	High collection efficiency	7	267 Dodges, combo x1.0
Clarity of goals/Rules	7	Clear objective: collect and avoid	7	Clear visual indicators and rules
Responsiveness of controls	6	Slight delay in response	7	Avg. reaction time: 0.01 ms
Pacing	7	Frequent actions and decisions	7	1801 Dashes in 30 s
Replayability	6	Random patterns ensure variety	6	Random patterns for enemies
Reward satisfaction	7	Points and survival bonuses	5	Collected 8 special items
Visual appeal	5	Clean minimalist design	6	Enhanced effects and hierarchy
Audio appeal	4	Basic sound feedback	5	Varied sound effects
Ease of learning	7	Simple mechanics and feedback	7	Instant feedback
Stress/Frustration level	0	Low tension, few near misses	5	Moderate stress from close calls
Sense of progress	7	Score progression	3	Only reached level 1
User interface clarity	6	Clear score and game-over display	6	Clear score and status displays

Table 4. Analysis of the results for the first version of the game (CircleDash 1.0) and suggestions for improvements.

Playing experience measure	Summary of analysis	Suggested improvements by cursor
Challenge level	Game was too easy	• Added dynamic difficulty scaling • Introduced "hunter" enemies • Increased enemy spawn rate and speed over time • Combo system for risk
Stress/Frustration	Lack of tension	• Near-miss detection for bonus points • Screen shake for high tension • More enemies at higher levels • Visual feedback and combos
Audio appeal	Limited variety	• New sounds: combos, near-misses, level-ups • Distinct sounds for special items • Audio cues for important events
Visual appeal	Basic visuals	• Dark theme + modern UI • Background grid for depth • Particle effects, pulsing special items • Pop-up scores and hunter enemy visuals
UI clarity/Responsiveness	Minor interface issues	• Difficulty indicator • Combo meter and multiplier • Tutorial text • Better contrast and button feedback • Clear visual hierarchy

recommendations, are presented in Table 4. Cursor identified that the game was perceived as too easy and lacking in emotional tension, which negatively affected scores related to challenge and stress/frustration. To address this, it proposed dynamic difficulty scaling, the introduction of "hunter" enemies, and a combo system that rewards risk-taking. Stress and excitement were further enhanced through mechanics such as near-miss detection, screen shake effects, and more intense enemy behavior at higher levels.

Regarding sensory feedback, the analysis revealed limited audio variety and basic visual design. In response, Cursor recommended adding distinct sound effects for combos, level-ups, and special items, as well as visual enhancements like a dark theme, modern UI elements, particle effects, and dynamic item visuals. For UI clarity and responsiveness, the feedback pointed to minor interface issues, leading to the inclusion of elements such as a difficulty indicator, combo meter, improved tutorial text, and clearer visual hierarchy. Overall, Cursor's suggestions aim to increase the game's emotional depth, sensory richness, and strategic complexity as key factors expected to enhance the AX scores and elevate the overall player experience.

4.2. Artificial experience (AX): Round 2

We asked Cursor to revise the game based on its analysis and the suggested improvements. This led to the development of a new version (CircleDash 2.0), where a snapshot of this version and a link to the playable game are shown in Figure 2. As reflected in the earlier analysis, the most prominent issue was the lack of challenge, followed by the simplicity of the user interface. To address these, several significant enhancements were made to improve player experience and increase replayability.

A Dynamic Difficulty system was introduced, enabling the game to adapt to the player's performance by progressively increasing the challenge through faster enemy movement, more frequent spawns, and the addition of "hunter" enemies. A Combo System was also implemented, rewarding players for quickly chaining collectible pickups under time pressure, with urgency signaled through visual and audio cues. This added a strategic element of risk versus reward to the gameplay. Additionally, Special Collectibles were added such as orange items worth triple points and marked with unique visuals to encourage more deliberate and tactical movement for score maximization. To enrich feedback and engagement, several Enhanced Feedback Mechanisms were integrated, including screen shake effects, score popups, near-miss detection, particle effects, and visual trails that provide a more dynamic and responsive feel. Finally, a Progressive Challenge structure was introduced, increasing the number and speed of enemies as players level up, while offering survival bonuses to reward skilled and sustained performance.

As in Round 1, we tasked Cursor with deploying a new AI agent to play CircleDash 2.0 and assess the gameplay experience using the same predefined playing experience measures. The results of this second evaluation are presented in Table 3. Compared to the first version, the scores show notable improvement, indicating that the changes significantly enhanced the AI player's experience. A key insight from this round is that many of the evaluation scores are grounded in quantitative gameplay data. For example, the high score for Engagement/Involvement is supported by a decision-making frequency of 1.3 decisions per second, while the score for Challenge Level reflects the intensity of gameplay, evidenced by the AI encountering six enemies and recording nine near-misses. This underscores a crucial distinction between Artificial Experience (AX) and User Experience (UX): whereas human players rely on subjective impressions, the AI evaluates gameplay based on concrete, recorded interactions with game elements. However, some scores are inherently expected regardless of gameplay. For instance, Clarity of Goals and Rules consistently receives a perfect score in AX evaluations, as the AI is preprogrammed with an understanding of the game's objectives and mechanics. This highlights a limitation of AX, where it is best suited for assessing experiential qualities that emerge through interaction, rather than structural features that can be inferred from the game's design.

Following this second evaluation, Cursor was again asked to analyze the results and propose further improvements. The outcomes of this analysis, along with the suggested design enhancements for CircleDash 2.0, are presented in Table 5.

4.3. Artificial experience (AX): Round 3

We asked Cursor to develop a third version of the game by incorporating the recommended features listed in Table 5. In the updated release (*CircleDash 3.0*), several significant improvements were introduced to deepen gameplay and enhance player engagement. A comprehensive achievement system was added, featuring goals such as performing 50 dashes in 10 s (Speedster), reaching level 5 (Survivor), achieving a 5x combo (Master Collector), executing 10 near misses in 5 s (Matrix Dodger), and surviving encounters with three hunter enemies (Hunter Slayer). The progression system now includes experience points earned through in-game actions like collecting items, surviving longer, and unlocking achievements. As players level up, the game dynamically increases in difficulty and rewards them with permanent power-ups.

Table 5. Analysis of the second version of the game (CircleDash 2.0) and suggestions for improvements.

Aspect	Score (1–7) initial issue	Improvements made
Challenge level	4/7 – Too easy	• Added "Berserker" enemies (erratic, larger) • Increased enemy spawn rates • Difficulty scales faster (every 15s) • Multiple simultaneous spawns • Enemies speed up more aggressively
Sense of progress	3/7 – Very poor progression	• Introduced XP-based leveling system • Visual XP bar • Rewards for level-ups • Achievement system with 5 challenges • Permanent upgrades unlocked through power-ups
Reward satisfaction	5/7 – Not rewarding enough	• Golden "jackpot" collectibles (5x points) • Achievement-based XP bonuses • Power-up system: Speed Boost, Shield, Point Multiplier • Visual/audio feedback for rewards
Stress level	5/7 – Could be more intense	• More aggressive enemy behavior • High-stakes Berserker enemies • Combo system with time pressure • Near-miss detection rewards • Tension via achievement challenges
Visual appeal	6/7 – Room for improvement	• Enhanced particle effects • Unique visuals for enemy types • Experience popups • Achievement and power-up notifications • Animated and more readable UI
Audio appeal	5/7 – Limited variety and feedback	• New sound effects for: achievements, level-ups • Varied sounds for collectibles • Improved audio balance • Context-sensitive audio feedback

Table 6. Analysis of the results for the third version of the game (CircleDash 3.0) and suggestions for improvements.

Feature	Observed problem	Improvement
Challenge level	Lacked gameplay variety	Added dynamic difficulty and goal-based achievements (e.g., near misses, hunter encounters).
Sense of progress	No clear progression	Introduced XP system, levels, and experience bar.
Reward satisfaction	Rewards felt minimal	Added permanent power-ups and visually distinct collectible multipliers.
Stress level	Low intensity	Included time-limited and high-stakes challenges.
Visual appeal	Plain visuals	Enhanced with effects for collectibles, UI upgrades, and visual cues.
Audio appeal	Missing or weak audio feedback	*Not addressed directly; could add audio cues for events, power-ups, and milestones.*

The collectibles system has also been enhanced with three distinct types including Normal (1x), Special (3x), and Jackpot (5x). They feature unique visual effects to aid recognition and encourage strategic targeting. A new power-up system introduces temporary enhancements such as Speed Boost, Shield, and Score Multiplier, some of which can become permanent. All power-ups are visually signaled to improve player feedback. Additionally, the user interface has been refined to improve usability and immersion. Key updates include an experience bar, real-time notifications for achievements and power-ups, a tutorial for onboarding new players, and an improved Game Over screen that summarizes performance metrics. Collectively, these changes make this version more rewarding, visually engaging, and strategically rich. A snapshot of *CircleDash 3.0* and the link to play this version are shown in Figure 2.

In Round 3, the final stage of the Artificial Experience (AX) evaluation, we asked Cursor to generate an AI player to simulate gameplay for 30 s and provide suggestions for improvements that are shown in Table 6.

4.4. Hypothetical experience (HX): Round 1

In the second approach proposed and tested in this study, rather than implementing the game and employing an AI player to simulate gameplay and generate Artificial Experience (AX), we focused on deriving feedback directly from the game design document. Specifically, we investigated the extent to which an AI model (GPT-4) could assess the gameplay experience based solely on the game's textual description. To this end, we utilized the original game design document introduced in the previous section (Table 2) and instructed $GPT-4$ tubo to act as a hypothetical player imagining a playthrough of the game. Following this simulated session, the AI was asked to evaluate the gameplay experience using the same set of player experience measures applied in the previous method. As with the AX approach, the AI assigned ratings on a $1\text{-to}-7$ Likert scale and provided brief justifications for each score. A summary of this evaluation is presented in Table 7.

According to the evaluation results presented in Table 7, the game offers a solid and engaging experience, particularly excelling in clarity of goals, ease of learning, and control responsiveness (each receiving the highest score of 7). The gameplay was rated as enjoyable and engaging (scores of 6), driven by the need for quick decision-making and a satisfying dash mechanic. Pacing and replayability were also positively received; however, the short session length slightly hindered the sense of flow and limited perceived progression, reflected in lower scores of 5 and 4, respectively. While the challenge level was considered well-balanced and enjoyable, the brief playtime restricted opportunities for deeper mastery. Visually and audibly, the game presented a clean and functional style, but further enhancements in feedback and reward systems could improve overall satisfaction and immersion. In summary, the design is well-suited for short, repeatable sessions but would benefit from more robust progression and reward mechanics to enhance long-term engagement.

Based on this analysis, we asked the AI to revise the game design to address the identified shortcomings. The result is a new version of the Game Design Document (GDD 2.0), summarized in Table 8. This updated design maintains the game's strengths such as intuitive one-touch controls, a fast and addictive core loop, and satisfying multisensory feedback across audio, visuals, and haptics, making it highly accessible and ideal for short play sessions. However, it also acknowledges the need for greater depth and sustained engagement. The revised design introduces improvements aimed at mitigating repetition, enhancing progression, and increasing opportunities for skill mastery. Although the esthetic

Table 7. Summary of the evaluations of the first version of the game design Document, along with justifications based on the scores provided by the AI model (GPT − 4 turbo).

Aspect	Score (1–7)	Reasons by AI agent playing the game for the given score
Fun/Enjoyment	6	The game provided quick, rewarding feedback and a satisfying dash mechanic. Collecting items was pleasing and incentivizing.
Engagement/Involvement	6	The need for fast decisions and reflex-based dashing kept me consistently involved during the 30 s.
Flow/Immersion	5	The rhythm of collecting and dodging encouraged immersion, though short session time and abrupt restarts slightly disrupted flow.
Challenge level	6	Enemy patterns required predictive movement and timing, offering meaningful challenge without being overwhelming.
Competence/Mastery	4	I could improve over time, but 30 s wasn't enough to fully master enemy patterns or optimize movement routes.
Clarity of goals and rules	7	Goals were immediately clear: collect, avoid, survive. Rules were visually and audibly reinforced through effects.
Responsiveness of controls	7	Tap-to-dash input was highly responsive, allowing precise control within the game's minimalist mechanics.
Pacing	6	The game maintained a brisk tempo, alternating between action and brief pauses after collisions. No slow or dead moments occurred.
Replayability	6	Unlockable cosmetic rewards and high score tracking gave incentives to replay. Short session length also encouraged retrying.
Reward satisfaction	5	Visual/audio cues for collecting points were pleasant. However, the reward system had limited depth within the first 30 s.
Visual appeal	5	Clean and minimalist. While not flashy, the visual clarity supported gameplay and reduced cognitive load.
Audio appeal	5	Simple sound effects provided adequate feedback. Audio cues were distinctive but minimalistic.
Ease of learning	7	Core mechanics were intuitive and understandable within the first few seconds. No tutorial needed.
Stress/Frustration level	3	Dying abruptly due to fast collisions introduced mild frustration, but immediate restarts helped reduce negative impact.
Sense of progress	4	Score increase and cosmetic unlocks created some progression, though not deeply felt within 30 s.
User Interface clarity	6	UI was minimal but effective—score display and restart flow were clear and distraction-free.

Table 8. The second version of the game design Document for CircleDash (GDD 2.0) generated by GPT − 4 turbo based on hypothetical experience.

Category	Suggested changes by GPT − 4 turbo on the game design document
Core mechanics	• Tap to dash – basic movement • Swipe to rotate – adds dash curve and skill • Collectibles: basic circles + "Fate Orbs" (slow time, multiplier, shield) • Adaptive Enemies – patrol, chase, area denial behaviors
Gameplay loop	• Dash and curve mid-air • Collect points and bonuses • Avoid increasingly complex enemies • Survive procedural waves • Earn XP to unlock story, cosmetics, modifiers
Game environments	• Biomes: neon grid, foggy temple, glitch realm • Unique music, filters, and enemy behaviors per biome • Biomes unlock by score or sequence
Visual & audio design	• Biome-specific particle effects • Dynamic audio that reacts to performance • Fate Orbs trigger slow-mo with custom effects
Progression system	• XP to level up and unlock passives (speed, shield, magnet) • Cosmetic unlocks (dash trails, themes) • "Story Scrolls" milestone-based narrative
Replayability enhancements	• Daily biome challenges (e.g., "no bounce," explosive enemies) • Endless and Story Modes • Achievements tracked in codex
Frustration mitigation	• Shield Orb = one free hit • Slow-moon near misses for reflex learning • Optional Assist Mode (global 20% slow)
Player skill curve	• Gradual introduction of enemies and mechanics • Biome-specific tactics • Bonus score for trick plays (e.g., close calls, multi-collects)
Scoring & feedback	• +1 Normal, +5 Fate Orbs • Bonus for time, biome goals • Streak = combo multiplier • Combo meter with glow and sound effects

style remains functional, it lacks strong emotional resonance. Additionally, the frustration caused by abrupt game-over moments could be reduced through supportive mechanics, and the inclusion of varied challenges or environments is suggested to help sustain player interest over time. The playable version of this game made by GPT − 4 turbo is available in https://sekhavat.itch.io/circlecollection-v2.

4.5. Hypothetical experience (HX): Round 2

In Round 2, the hypothetical AI agent was tasked with evaluating GDD 2.0, which had been developed based on the feedback from the previous round. In this evaluation, the AI assessed the gameplay experience solely by interpreting the revised design document, simulating a playthrough of the hypothetical game rather than interacting with an actual implementation. The resulting Hypothetical Experience (HX) scores for GDD 2.0 are presented in Table 9. Additionally, we instructed GPT − 4 Turbo to generate a playable prototype of the game based on this updated design.

The evaluation indicates that the game delivers simple, accessible, and fast-paced gameplay, with clear objectives (Clarity of Goals and Rules: 7) and fairly responsive controls (Responsiveness: 6), making it well-suited for casual players. Fun and pacing are rated moderately (both 5), driven by quick, reflex-based interactions and straightforward physics. However, the game lacks depth and long-term engagement—low scores in Reward Satisfaction (3), Sense of Progress (3), and Replayability (4) suggest limited motivation for continued play. While aspects like engagement, challenge, and immersion are average (scores of 4–5), repetitive patterns and a lack of evolving content hinder sustained interest. Visually and audibly, the game performs adequately but lacks distinctive appeal (Visual Appeal: 4, Audio Appeal: 5). Overall, it's a solid casual experience that could be significantly enhanced with more variety, progression systems, and richer emotional or esthetic elements.

Based on this analysis, we asked the AI to revise the game design to address the identified shortcomings. The result is a new version of the Game Design Document (GDD 3.0), summarized in Table 10. The playable version of this game made by GPT − 4 turbo is available in https://sekhavat.itch.io/circle-collection-v3.

Table 9. Summary of the evaluations of the second version of the game design document (GDD 2.0), along with justifications based on the scores provided by the AI model (GPT − 4 turbo).

Parameter	Score (1–7)	Evaluation
Fun/ Enjoyment	5	The core loop is fast and reflex-based, which can be enjoyable in short bursts. The simplicity and bounce physics add to the fun.
Engagement/Involvement	4	Somewhat engaging due to fast decisions and dodging, but repetition may reduce long-term involvement.
Flow/Immersion	4	The player can enter a flow state during gameplay, but minimal narrative or evolving mechanics may limit immersion.
Challenge level	5	Predictable enemy patterns and timing-based movement offer decent challenge, but could become too easy or repetitive over time.
Competence/Mastery	4	Some sense of improving timing and movement, but limited depth reduces sense of mastery.
Clarity of goals and rules	7	Very clear: avoid enemies, collect points, survive. Rules are easy to grasp instantly.
Responsiveness of controls	6	One-touch control should be very responsive, assuming well-implemented input and physics.
Pacing	5	Quick rounds and fast movement support solid pacing. Could benefit from progressive difficulty or tempo changes.
Replayability	4	Unlockables help, but limited variation may hinder long-term replay value.
Reward satisfaction	3	Basic score-based rewards and cosmetic unlocks are simple but not deeply satisfying.
Visual appeal	4	Minimalist style can be effective, but may lack visual excitement or personality.
Audio appeal	5	Simple feedback sounds help create polish, though music or richer effects would improve atmosphere.
Ease of learning	7	Extremely easy to learn; perfect for casual players.
Stress/Frustration level	3	Might feel unfair or repetitive without dynamic difficulty or player control in movement direction.
Sense of progress	3	Progress only via score and unlocks—no level changes or new mechanics to mark advancement.
User interface clarity	6	Presumably simple UI matches game type; score and feedback are clear.

Table 10. The third version of the game design Document for CircleDash (GDD 3.0) generated by GPT − 4 turbo based on hypothetical experience.

Category	Details
Game mechanics	• Directional dash: Tap + Swipe for controlled direction. • Combo meter: Bonuses for rapid collection. • Power-ups: Invincibility, speed boost, enemy freeze. • Enemy variety: Homing, pulsating, and teleporting enemies introduced progressively.
Gameplay loop	• Swipe-tap to dash. • Chain circles for combos. • Use power-ups. • Survive increasingly tough waves. • Unlock rewards and visual upgrades.
Reward & progression	• Dynamic progression: Arena and enemies evolve over time (30s, 60s, etc.). • Tiered unlockables: Shapes, effects, power-ups. • Achievements: Survival, collection, streak challenges.
Enhanced feedback	• Soundtrack: Music intensifies as you progress. • SFX: Rich, varied sounds for all actions. • Haptics: Vibration feedback for events like bounce, danger, and power-ups.
Visual & UI enhancements	• Visuals: Neon glow, particles, shifting backgrounds. • UI: Timer, combo meter, and power-up indicators.

Table 11. Summary of the evaluations of the third version of the game design document (GDD 3.0), along with justifications based on the scores provided by the AI model (GPT − 4 turbo).

Parameter	Score (1–7)	Evaluation
Fun/Enjoyment	5	Simple and fun, but can become repetitive.
Engagement/Involvement	4	Limited long-term engagement without variety or progression.
Flow/Immersion	4	Intermittent flow due to abrupt game-overs and static environment.
Challenge level	5	Offers reflex-based challenge, but can become predictable.
Competence/Mastery	4	Players can improve, but lacks advanced techniques or skills.
Clarity of goals and rules	7	Clear and easily understandable rules and goals.
Responsiveness of controls	7	One-tap input is very responsive.
Pacing	4	Pacing is uniform, lacking intensity or escalation.
Replayability	4	Replayable in short bursts, but lacks incentives for long-term play.
Reward satisfaction	3	Cosmetic unlocks only; feels superficial over time.
Visual appeal	4	Clean and minimalistic, but lacks variety and polish.
Audio appeal	3	Basic sounds; could benefit from music and layered audio feedback.
Ease of learning	7	Extremely easy to learn.
Stress/Frustration level	5	Fair balance, but sudden game-over may frustrate some.
Sense of progress	3	No levels, achievements, or score-based unlocks beyond visuals.
User interface clarity	6	UI is minimal and functional.

4.6. Hypothetical experience (HX): Round 3

During Round 3, the AI model was asked to analyze GDD 3.0, which had been revised in response to feedback from the initial evaluation. Instead of playing a working version of the game, the AI inferred the gameplay experience by mentally simulating how the game would play out based on the updated design document. The outcomes of this imagined gameplay session referred to as Hypothetical Experience (HX) are summarized in Table 11. Furthermore, GPT − 4 turbo was used to transform the updated game design into a functional prototype.

The game was found to be simple and enjoyable (score: 5), with clear goals and intuitive controls (scores: 7 for both clarity and responsiveness), making it extremely easy to learn (score: 7). However, the experience suffers from limited depth and variety, resulting in only moderate engagement, flow, and replayability (scores: 4). While the reflex-based challenge is fair (score: 5), the lack of advanced skills, meaningful progression, and rewarding feedback reduces long-term appeal. Esthetic elements like visuals and audio were rated as basic (scores: 4 and 3, respectively), and reward satisfaction was notably low (score: 3), pointing to the need for more engaging and layered incentives. Overall, the game is accessible and functional for short play sessions but would benefit from improvements in progression, audiovisual depth, and motivational structure.

Following this evaluation, the AI was prompted to redesign the game in order to resolve the issues highlighted in the analysis. This led to the creation of an updated Game Design Document, referred to as GDD 4.0, which is outlined in Table 12.

Table 12. The fourth version of the game design document for CircleDash (GDD 4.0) generated by GPT − 4 turbo based on hypothetical experience.

Category	Details
Game mechanics	• Tap/Click to dash: Directional dash • Bounce physics: Rebound off arena walls • Power circles: Temporary effects • Combo timer: Score multipliers from quick successive collects
Gameplay loop	• Dash to navigate the arena • Collect circles, avoid enemies • Activate power-ups and score combos • Progress through harder levels • Unlock achievements and upgrades
Game objects	• Player circle: Neon style, customizable • Enemy circles: Pattern-based movement • Power circles: Temporary bonuses (e.g., slow-mo) • Walls: bounce with glow pulse- Combo Gauge: Tracks streaks
Rules	• +1 Point per collectible • Lose a life on enemy contact (3 lives) • Combo = 3+ collectibles in <5s • Power-ups last 5–10s • New enemy types and visuals unlock through progression
Environment	• Neon Arena with dynamic background • Theme shifts by milestone (e.g., blue→red) • Particle effects on events (collect, crash)
Player dynamics	• Simple control scheme • Mastery through timing and direction • Higher skill needed in advanced levels
Reward system	• Score unlocks: Trails, skins, arenas • Missions: Daily/weekly goals • Stars for upgrades (e.g., power-up duration)
Feedback	• Visual: Glows, trails, pulses • Audio: Synthwave music, SFX (ping, buzz) • Haptics: Light vibration on mobile
Scoring system	• +1 Per collectible • +1 Per 10s survived • Combo bonus • Level-clear bonuses • Display: Score, combo, lives, high score
UI & UX enhancements	• HUD: Lives, combo bar, level indicator • Pause/Menu access via tap • Colorblind Mode toggle in settings

5. Discussion and conclusion

The comparative evaluation of Artificial Experience (AX) and Hypothetical Experience (HX) techniques reveals important differences in their contributions to the iterative game design process. Both methods demonstrate the promise of leveraging AI to support user experience evaluation; however, they operate at fundamentally different levels of abstraction and generate distinct types of insights. HX relies on the AI's interpretation of the game design document without direct interaction with the game. This approach provides a relatively efficient and accessible means of identifying design issues early in development by delivering structured feedback across various UX dimensions without requiring a fully functional prototype. Nevertheless, our findings indicate that although HX can suggest useful improvements, these recommendations often result in superficial or inconsistent changes.

The limitation likely stems from the static and interpretive nature of HX, where the AI simulates an imagined player experience based on documentation. By combining HX with AX, which involves AI agents interacting directly with the game, the proposed approach enables the use of both hypothetical and artificial players to evaluate and refine games during early development stages, providing actionable feedback before extensive human playtesting is required. In contrast, AX engages an AI agent in actively playing the game, grounding its evaluations in gameplay data and in-game behaviors. This approach consistently yields more targeted, evidence-based design revisions that improve the user experience across multiple dimensions without undermining previously strong aspects. For example, changes informed by HX occasionally reduced scores in measures that were previously high, whereas AX-driven iterations enhanced weaker areas while preserving existing strengths. This demonstrates the focused and context-sensitive quality of AX. The success of AX also underscores its ability to capture experiential factors that are difficult to infer from design documents alone, such as real-time pacing, flow, and emergent difficulty curves. The integration of quantitative gameplay metrics, such as decision frequency and threat encounters, enriches AX's feedback with data-driven insights, making it particularly suitable for refining interactive qualities that are inherently dynamic and nonlinear.

Both approaches struggled to fully address progression and emotional depth during the early evaluation rounds. While HX identified these as concerns, it failed to propose systemic solutions. By contrast, AX encouraged the inclusion of dynamic difficulty adjustments, layered collectibles, achievement systems, and more nuanced feedback mechanisms, all of which led to measurable gains in engagement, mastery, and reward satisfaction in subsequent iterations. These developments demonstrate AX's capacity to support iterative feedback loops that evolve not only superficial features but also deeper structural mechanics within gameplay. However, AX is not without its limitations. Certain UX aspects, such as clarity of goals and user interface elements, were consistently rated higher by AX than might be expected from a human player's perspective. This discrepancy likely arises because the AI has privileged access to internal game logic and mechanics, which human players must discover through interaction and learning. Consequently, AX may not fully replicate the cognitive challenges or learning curves experienced by real players. To mitigate this, AX is most effective when combined with human testing or when configured to simulate less omniscient AI agents that better approximate human behavior. By introducing variability in the agents' decision-making, reaction times, or exploration strategies, AX can provide more realistic feedback on potential usability issues and gameplay challenges, making it a valuable tool for early-stage evaluation and iterative game design.

Overall, AX emerged as a more robust and insightful tool for diagnosing design weaknesses and guiding systemic improvements. HX remains valuable as a low-cost heuristic during early design ideation but lacks the depth and consistency necessary for high-fidelity iterative development. The findings suggest a complementary evaluation framework where HX vets initial concepts and AX refines them through simulated, data-informed experiential feedback. Implementing this dual strategy within an AI-driven development pipeline holds promise for accelerating and enhancing the game design process in both professional and educational settings. Despite these promising results, several limitations should be acknowledged. This study focused on a minimalist arcade-style game, CircleDash, which features straightforward mechanics and limited strategic complexity. As such, the generalizability of AX and HX to more intricate genres involving layered mechanics, narrative elements, or long-term decision-making remains uncertain. Future research should apply these methods to more complex games, including role-playing, strategy, or multiplayer titles, to assess their ability to capture emergent gameplay dynamics and richer player experiences.

Additionally, a mid-skill AI agent was employed to standardize gameplay simulations in AX. While this approach maintained consistency, it restricted the analysis to a single player skill level. Exploring how AX performs across different simulated player competencies, such as novice or expert levels, could provide deeper insight into how design choices affect various player types. This avenue offers fertile ground for future investigation. Lastly, this study did not incorporate data from real human players as a benchmark. Integrating traditional user testing alongside AX and HX evaluations would provide valuable validation for these AI-driven methods. Mixed-method research comparing AI-generated feedback with human player responses could illuminate areas of alignment and divergence, thereby strengthening the reliability and applicability of AX and HX as evaluation frameworks. Together, these limitations highlight significant opportunities for advancing AI-powered tools that enable scalable, adaptive, and nuanced user experience design in games.

Acknowledgment

The author affirm that GPT − 4 Turbo is used exclusively for grammar checking and language refinement of this article. The content, analysis, and revisions of the manuscript were entirely conceived, written, and finalized by the author. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Yoones A. Sekhavat**: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

ORCID

Yoones A. Sekhavat http://orcid.org/0000-0003-3654-9583

References

Anjum, A., Li, Y., Law, N., Charity, M., & Togelius, J. (2024, May). The ink splotch effect: A case study on chatgpt as a co-creative game designer. In G. Smith, J. Whitehead, B. Samuel, K. Spiel, & R. van Rozen (Eds.), *Proceedings of the 19th International Conference on the Foundations of Digital Games* (pp. 1–15). Association for Computing Machinery (ACM).

Baidoo-Anu, D., & Ansah, L. O. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AI*, *7*(1), 52–62. http://doi.org/10.2139/ssrn.4337484

Barthet, M., Khalifa, A., Liapis, A., & Yannakakis, G. (2022, September). Generative personas that behave and experience like humans. In K. Karpouzis, S. Gualeni, J. Pirker, & A. Fowler (Eds.), *Proceedings of the 17th International Conference on the Foundations of Digital Games* (pp. 1–10). Association for Computing Machinery (ACM).

Caroux, L., & Pujol, M. (2024). Player enjoyment in video games: A systematic review and meta-analysis of the effects of game design choices. *International Journal of Human–Computer Interaction*, *40*(16), 4227–4238. https://doi.org/10.1080/10447318.2023.2210880

Choi, J. O., Forlizzi, J., Christel, M., Moeller, R., Bates, M., & Hammer, J. (2016, October). Playtesting with a purpose. In A. L. Cox, P. O. Toups Dugas, R. L. Mandryk, P. A. Cairns, V. Vanden Abeele, & D. M. Johnson (Eds.), *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play* (pp. 254–265). Association for Computing Machinery (ACM).

Colado, I. J. P., Colado, V. M. P., Morata, A. C., Píriz, R. S. C., & Manjón, B. F. (2023, October). *Using new AI-driven techniques to ease serious games authoring* [Paper presentation]. 2023 IEEE Frontiers in Education Conference (FIE), College Station, TX (pp. 1–9). IEEE. https://doi.org/10.1109/FIE58773.2023.10343021

Cook, M. (2020, August). *Software engineering for automated game design* [Paper presentation]. 2020 IEEE Conference on Games (CoG), Osaka, Japan (pp. 487–494). IEEE. https://doi.org/10.1109/CoG47356.2020.9231750

Fan, M., Yang, X., Yu, T., Liao, Q. V., & Zhao, J. (2022). Human-AI collaboration for UX evaluation: Effects of explanation and synchronization. In S. Bardzell, S. Lindley, A. Sarcevic, H. Kuzuoka, K. Reinecke, H.-C. Wang, & N. Yamashita (Eds.), *Proceedings of the ACM on Human-Computer Interaction, 6(CSCW1)* (pp. 1–32). Association for Computing Machinery.

Fui-Hoon Nah, F., Zheng, R., Cai, J., Siau, K., & Chen, L. (2023). Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. *Journal of Information Technology Case and Application Research*, *25*(3), 277–304. https://doi.org/10.1080/15228053.2023.2233814

Fullerton, T. (2024). *Game design workshop: A playcentric approach to creating innovative games*. AK Peters/CrC Press.

Hu, X., Liu, A., Li, X., Dai, Y., & Nakao, M. (2023). Explainable AI for customer segmentation in product development. *CIRP Annals*, *72*(1), 89–92. https://doi.org/10.1016/j.cirp.2023.03.004

Huang, Y., Kanij, T., Madugalla, A., Mahajan, S., Arora, C., & Grundy, J. (2024). Unlocking adaptive user experience with generative AI. In H. Kaindl, M. Mannion, & L. Maciaszek (Eds.), *Proceedings of the 19th International Conference on Evaluation of Novel Approaches to Software Engineering* (Vol. 1, pp. 760–768). Scitepress. https://doi.org/10.5220/0012741000003687

Isbister, K., & Schaffer, N. (2008). *Game usability: Advancing the player experience*. CRC press.

Jang, W., Zhuo, G., Pyun, H., & Lee, G. (2025). Competing with artificial intelligence or other fans: Effects of making predictions about game outcome on fans' perceived curiosity and evaluations of game consumption. *International Journal of Human–Computer Interaction*, *41*(1), 765–774. https://doi.org/10.1080/10447318.2024.2327178

Jiang, E., Olson, K., Toh, E., Molina, A., Donsbach, A., Terry, M., & Cai, C. J. (2022, April). *Promptmaker: Prompt-based prototyping with large language models* [Paper presentation]. CHI Conference on Human Factors in Computing Systems Extended Abstracts, New Orleans, LA (pp. 1–8).

Koster, R. (2013). *Theory of fun for game design*. O'Reilly Media, Inc.

Kreminski, M., Dickinson, M., Osborn, J., Summerville, A., Mateas, M., & Wardrip-Fruin, N. (2020, October). Germinate: A mixed-initiative casual creator for rhetorical games. In L. Lelis & D. Thue (Eds.), *Proceedings of the Sixteenth Artificial Intelligence and Interactive Digital Entertainment Conference (AIIDE 20)* (Vol. 16, No. 1, pp. 102–108). AAAI Press.

Kumaran, V., Rowe, J., Mott, B., & Lester, J. (2023, October). Scenecraft: Automating interactive narrative scene generation in digital games with large language models. In M. Eger & R. E. Cardona Rivera (Eds.), *Proceedings of the Nineteenth AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment* (Vol. 19, No. 1, pp. 86–96). AAAI Press.

Liapis, A., Yannakakis, G. N., & Togelius, J. (2013). Sentient sketchbook: Computer-assisted game level authoring. In G. N. Yannakakis, E. Aarseth, K. Jørgensen, & J. C. Lester (Eds.), *Proceedings of the 8th International Conference on the Foundations of Digital Games* (pp. 213–220). Society for the Advancement of the Science of Digital Games.

Luo, Y. (2025). Designing with AI: A systematic literature review on the use, development, and perception of AI-enabled UX design tools. *Advances in Human-Computer Interaction*, *2025*(1), 3869207. https://doi.org/10.1155/ahci/3869207

Luo, Y., Gkouskos, D., Russo, N. L., & Wang, M. (2024). Navigating from data-driven design to designing with ML: A case study of truck HMI system design. In M. Štorga, S. Škec, T. Martinec, D. Marjanović, & N. Pavković (Eds.), *Proceedings of the DESIGN 2024 – 18th International Design Conference* (Vol. 4, pp. 2119–2128). Cambridge University Press. https://doi.org/10.1017/pds.2024.214

Lv, Z. (2023). Generative artificial intelligence in the metaverse era. *Cognitive Robotics*, *3*(1), 208–217. https://doi.org/10.1016/j.cogr.2023.06.001

Meşe, C., Aydemir, H., Solak, M. Ş., Gök, M., & Bolat, Y. İ. (2025). Digital game designers' experiences in developing narrative design based on value-sensitive design. *International Journal of Human–Computer Interaction*, *41*(7), 1–14. https://doi.org/10.1080/10447318.2025.2531282

Mirza-Babaei, P., Moosajee, N., & Drenikow, B. (2016, October). Playtesting for indie studios. In M. Turunen, S. Kotilainen, S. Rubio Hernandez, J. Tuominen & T. Väliharju (Eds.), *Proceedings of the 20th International Academic Mindtrek Conference* (pp. 366–374). Association for Computing Machinery (ACM).

Mirza-Babaei, P., Moosajee, N., & Drenikow, B. (2016b, May). Games user research (GUR) for indie studios. In J. Kaye & A. Druin (Eds.), *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (pp. 3159–3165). Association for Computing Machinery.

Mirza-Babaei, P., Stahlke, S., Wallner, G., & Nova, A. (2020, April). A postmortem on playtesting: Exploring the impact of playtesting on the critical reception of video games. In J. Y. C. Chen & G. F. Ko (Eds.), *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (pp. 1–12). Association for Computing Machinery. https://doi.org/10.1145/3313831.3376831

Moon, J., Lee, U., Koh, J., Jeong, Y., Lee, Y., Byun, G., & Lim, J. (2024). Generative artificial intelligence in educational game design: Nuanced challenges, design implications, and future research. *Technology, Knowledge and Learning*, *30*(1), 447–459. https://doi.org/10.1007/s10758-024-09756-z

Ratican, J., & Hutson, J. (2024). Adaptive worlds: Generative AI in game design and future of gaming, and interactive media. *ISRG Journal of Arts, Humanities and Social Sciences*, *2*(5), 284–290. https://doi.org/10.5281/zenodo.13894497

Schell, J. (2008). *The art of game design: A book of lenses*. CRC press.

Schoop, E., Zhou, X., Li, G., Chen, Z., Hartmann, B., & Li, Y. (2022, April). Predicting and explaining mobile UI tappability with vision modeling and saliency analysis. In S. Barbosa, C. Lampe, C. Appert, D. A. Shamma, S. Drucker, J. Williamson, & K. Yatani (Eds.), *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1–21). Association for Computing Machinery. https://doi.org/10.1145/3491102.3517497

Sekhavat, Y. A., Mani, J., Mostafavi, S. V., Roohi, S., Movahedi, Y., & Zhen, L. (2025). Safe walk: A serious game for exploring environmental distractions affecting pedestrian safety. *International Journal of Serious Games*, *12*(3), 81–97. https://doi.org/10.17083/02xf3365

Sekhavat, Y., & Tabatabaei, A. (2025). Coming together: Exploring predestined interactions through an interactive sculpture inspired by the Adam and Eve story. *International Journal of Performance Arts and Digital Media*, *21*(3), 1–25. https://doi.org/10.1080/14794713.2025.2568298

Sekhavat, Y. A., & Zarei, H. (2018). Sense of immersion in computer games using single and stereoscopic augmented reality. *International Journal of Human–Computer Interaction*, *34*(2), 187–194. https://doi.org/10.1080/10447318.2017.1340229

Takaffoli, M., Li, S., & Mäkelä, V. (2024, July). Generative AI in user experience design and research: How do UX practitioners, teams, and companies use GenAI in industry? In A. Vallgårda, L. Jönsson, J. Fritsch, S. Fdili Alaoui, & C. A. Le Dantec (Eds.), *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS 2024)* (pp. 1579–1593). Association for Computing Machinery. https://doi.org/10.1145/3643834.3660720

van Rozen, R. (2025, April). Live game design: Prototyping at the speed of play. In J. Pirker, F. Kayali, K. Spiel, S. Harrer, A. Khalifa, & G. A. B. Barros (Eds.), *Proceedings of the 20th International Conference on the Foundations of Digital Games* (FDG '25) (pp. 1–12). Association for Computing Machinery. https://doi.org/10.1145/3723498.3723726

Weber, I. (2025). Large language models as software components: A taxonomy for LLM-integrated applications. arXiv Preprint arXiv:2406.10300.

Yang, D., Kleinman, E., & Harteveld, C. (2025). GPT for games: An updated scoping review (2020–2024). *IEEE Transactions on Games*. Advance online publication. https://doi.org/10.1109/TG.2025.3563780

About the author

Yoones A. Sekhavat is an Assistant Professor at the Modern College of Business and Science, Oman, and former lecturer at Uppsala University, Sweden. He holds a PhD in Computer Science from Memorial University, Canada, and researches HCI, serious games, and AI-driven creativity.